Rudradev Arya

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Education

MIT-ADT, Pune **Nov. 2021 – June 2025**
B.Tech in Computer Science and Engneering, Specializing in Cloud Computing *Pune, MH*

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Operating Systems
- Systems Programming
- Software Methodology
- Database Management
- Internet of Things
- Cloud Fundamentals

Projects

LambdaMailer | *Node.js, Amazon Lambda, Amazon SES* **February 2024**

- Leveraged **AWS Beanstalk** for scalable backend deployment and **Vercel** for frontend with which users can upload a CSV with relevant target mail IDs and other information
- Acheived a success rate of **99.2%** for **50,000** emails at just **$40/month** at maximum throughput of **200 mails/sec** and at maximum capacity
- Utilized **React** and **React Router** for a user friendly interface and **OAuth 2.0** protocol for secure authentication and authorization and minimal
- Designed the project as a microservice, communicating via APIs built with **Axios** and **Express.js**

EventU | *Next.js, Appwrite, TypeScript* **August 2023**

- Engineered a complex event management system leveraging **NoSQL** databases; streamlined unstructured data processing and achieved a **40%** reduction in data retrieval time, enhancing overall event coordination efficiency
- Designed the user interface using HTML, **TailwindCSS**, and **Next.js**, enhancing the platform's usability.
- Leveraged **Appwrite** for authentication, multiple sign-in methods, and efficient storage and file management.
- Utilized Next.js, an open-source React-based framework, to achieve server-side rendering and enhanced **SEO**. Deployed the application on **Vercel's** frontend cloud infrastructure for optimal performance.

Elevator Waiting Time Estimation and Simulation | *Arduino PoC, Python, Java, C++* and **Jan 2023**

- Implemented Various **Algorithms** using **Java** like Round Robin, Up Peak, Zoning etc to effectively simulate **Scheduling Algorithms** in an elevator
- Created a **Proof Of Concept(PoC)** model using Arduino and C++ to demonstrate real world effectiveness of Scheduling algorithms and to effectively **estimate** waiting time for an elevator

Minecraft Replica using Python | *Python, Pyglet, Linear Algebra* **April 2023**

- Utilized **Pyglet** a Python multimedia library to create a **128x128** Block replica of the original Minecraft game
- Leverage basic **Linear Algebra** to understand various game development techniques such as cursor and player movement correlation
- Implemented **object-oriented programming** practices such as abstraction to abstract away user controls and player movements
- Implemented **CUDA** to parallelize the workload of the game improving the FPS from **50 FPS** on CPU to over **270 FPS** on a mid tier GPU

Technical Skills

Languages: Python, C/C++, JavaScript, SQL, TypeScript, Node.js, CUDA
Developer Tools: VS Code, Git, SVN, Jupyter NoteBook
Cloud Technologies: GCP, AWS, Amazon RDS, S3, Lambda, SES, SQS
Web Frameworks: React, Next.js, Express,js
Database Systems: MongoDB, Firebase, Amazon RDS
Deployment and Monitoring: AWS Elastic Beanstalk, Postman, Vercel

Certifications

Nvidia : Fundamentals of Accelerated Computing with CUDA C/C++
AWS Cloud Quest: Cloud Practitioner
AWS Academy Graduate : AWS Academy Cloud Architecting

Achievements

- Solved more than **400**+ on various coding platforms like **Codeforces** and **GeeksForGeeks**.
- Participated and **Won** in ValiCQ, university level coding competition hosted by **CodeChef-MITSOE** Chapter.
- Participated at national level **GPU** Computing **Codeathon** organized by NVIDIA DLI.
- Top 30 on the University level GeeksForGeeks Leaderboard and Improving.